Option Pricing For Weather-Based Derivatives

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March 20, 2026

Motivation

Goal:

Model call/put options based on weather for weather-sensitive industries

Air Travel

Agriculture

Tourism

Ornstein–Uhlenbeck Process [Gib24]

Gauss-Markov process defined by SPDE:

Mean-reverting OU-Process

$$dT_t = \kappa(\overline{T}_t - T_t)dt + \sigma_t dW_t$$

- T_t = Daily average temperature at time t
- $\overline{T}_t$ = mean temperature at time t
- κ = "constant"
- σ_t = volatility at time t
- W_t = Wiener process

Explicit Solution

$$T_t = \overline{T}_t + \exp\left(-\int_0^t \kappa du\right) \int_0^t \exp\left(\int_0^t \kappa du\right) \sigma_t dW_t$$

Call/Put Pricing [Gib24]

Degree Days (Heating/Cooling)

- $CDD_n = (T_n - 65)^+, \qquad n \in$ Nov 15, ..., Mar 15
- $HDD_n = (65 - T_n)^+, \qquad n \in$ May 15, ..., Oct 15
- $DD_N = \sum_{n=1}^{N} CDD_n$ (or HDD_n)

Let K_c = strike price, α = payout rate, C = cap price. Then

- Call price $= \min\left\{\alpha(DD - K_c)^+, C\right\}$
- Put price $= \min\left\{\alpha(K_c - D)^+, C\right\}$

Typical values are $2.5K \leq \alpha \leq 5K$, $0.5M \leq C \leq 1M$

Application

1. Use hourly dry bulb temperature from Boston Logan Airport in 2012 [26]
2. Estimate daily average using trapezoidal integration
3. Fit daily average temperature with quadratic regression
4. Compute ACF and PACF of residuals
5. Estimate σ_t by using a monthly average and linearly interpolate to get daily values
6. Estimate T_t using OH equation with $\kappa \equiv 1$
7. Compute call and put options based on modeled temperature
8. Run MC simulations

References

[26] 2026. URL: https://www.ncei.noaa.gov/access/search/index.

[Gib24] Jorge C Gibert. "Monte Carlo Pricing of Temperature Weather Derivatives". In: *Department of Mathematics and Statistics, Florida International University* (2024), p. 1.